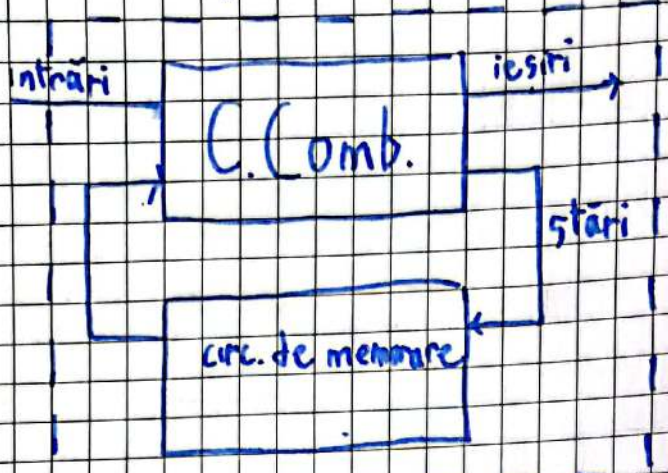


Circuite seventiale

istoric functional

26
00



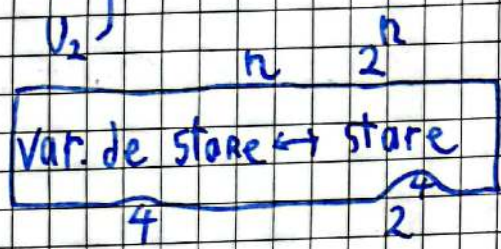
$$F(u, x) \rightarrow (x, y)$$

F_1
 F_2
 u_1
 u_2

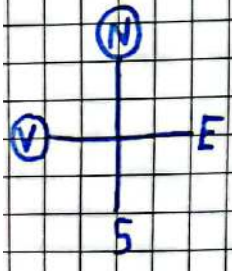
$u_1 = \hat{1}$
 $u_2 = \hat{1}$

$F_1 = \hat{1}$
 $F_2 = \hat{1}$

$\Rightarrow$ stare
 var. de stare



automat de intersetie



stare 1: N-S
 stare 2: V-E
 stare 3: pauză

var. de stare
 2 var.



starea 0

tranzitie $\rightarrow$



cinc asincrone

-||- / -||-

?tren de impulsuri?

Def.: automat finit

$$A = \{U, X, Y, f, g\}$$

cuvintu plu ordonat

U - multimea var. de intrare

X - finita avok de stare

Y - semnalelor de iesire

f: $X \times U \rightarrow X$ func. de tranzitie

g: $X \times U \rightarrow Y$ func. de iesire

ASM algoritmic state machine

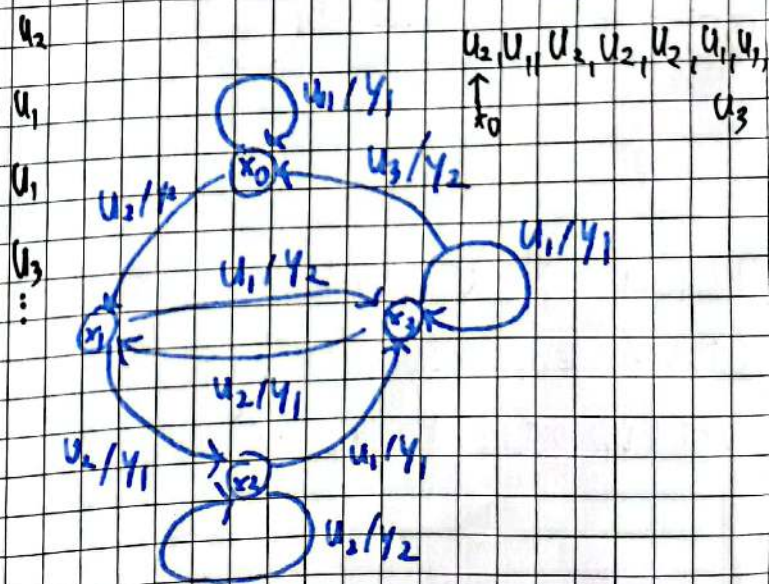
u_2	x_0	y_1
u_1	x_1	y_2
u_2	x_2	y_1
u_2	x_1	y_1

Circuite Mealy

1^a - stare initială x

2^a - secventa de

intrări



$$x: \{x_0, x_1, x_2, x_3\}$$

$$u: \{u_1, u_2, u_3\}$$

$$y: \{y_1, y_2\}$$

$T: f, g$ conf. figurii

Tema

$$\text{Fie } U = \{u_1, u_2\}$$

$$x = \{x_1, x_2, x_3\}$$

$$Y = \{y_1, y_2\}$$

Avem urm. func. Să se deseneze graf

$$f(x_1, u_1) = x_2$$

$$f(x_1, u_2) = x_3$$

$$f(x_2, u_1) = x_1$$

$$f(x_2, u_2) = x_3$$

$$f(x_3, u_1) = x_2$$

$$f(x_3, u_2) = x_3$$

$$g(x_1, u_1) = y_1$$

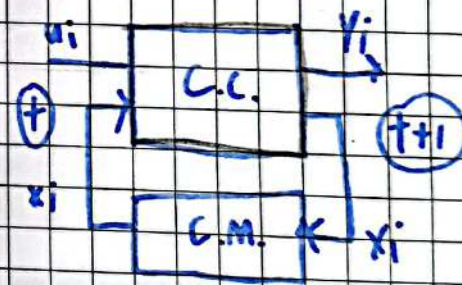
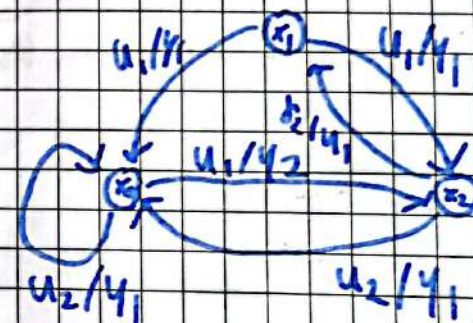
$$g(x_1, u_2) = y_2$$

$$g(x_2, u_1) = y_2$$

$$g(x_2, u_2) = y_1$$

$$g(x_3, u_1) = y_2$$

$$g(x_3, u_2) = y_1$$

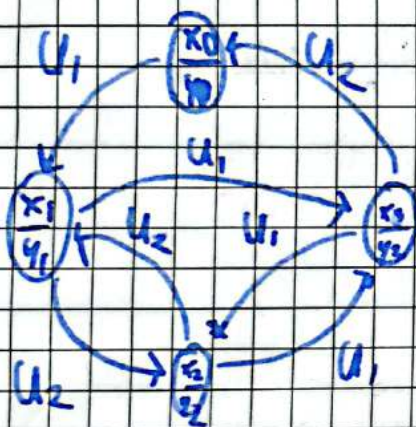
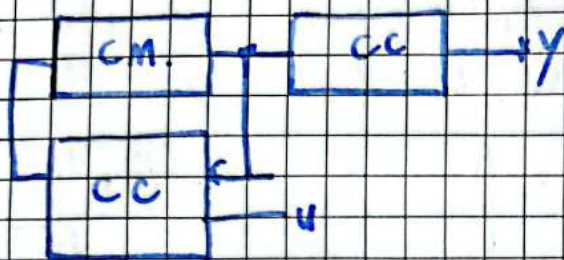


Moore

În automat ieșirile depind exclusiv de stare (nu și de intrări)

$$y^t = f(x_i^t)$$

$$x_i^{t+1} = f(x_i^t, u_i^t)$$



$$g(x_0) = y_1$$

$$g(x_1) = y_1$$

$$g(x_2) = y_2$$

$$g(x_3) = y_2$$

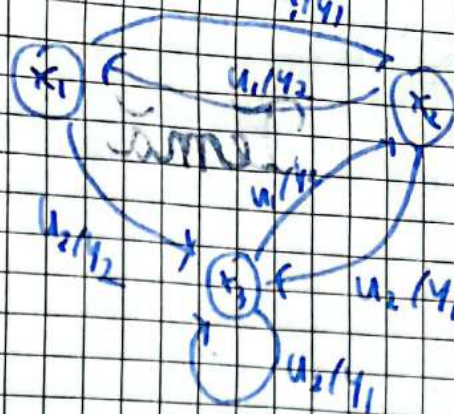
Se înlocuiește fiecare

stare a automatului Milie cu atâtea

stări ~~care~~ sunt asociate câte ieșiri

sunt asociate tranzițiile care intră

în acea stare



Mealy

x1

x2

x3

Moore

x1/y2

x2/y1

x2'/y2

x3/y1

x3'/y2

